

CONJURA OPEN PROBLEM

Parallel Presampling

From split auxiliary input to split bit-fixing

Status. Statement: AI-written, not yet formalized. Open at every $\ell \geq 2$. The case of deterministic preprocessing is immediate from [CDGS] (Remark 6), so the whole content lies in the private coins. This document states the conjecture and nothing else; what follows from it, with proofs, is [PP].

Category. *Idealized Models & Non-Uniformity.*

Conjecture (informal). *Several parties read an entire random function, without communicating and on independent coins, and each writes down a short string. An online adversary is then given all the strings and a bounded number of queries to the function. The conjecture says that each party can, on its own, mark a short list of points of the function, so that the online adversary cannot tell the real function from one fixed in advance on all the lists together and fresh everywhere else. Because the parties mark points of the same function, no two lists disagree where they overlap, so no arbitration is needed.*

1 The setting

Presampling is the standard route to non-uniform security in idealized models. A random oracle together with S bits of oracle-dependent advice is close to a mixture of oracles fixed on P points and uniform elsewhere, at a cost of roughly ST/P against T -query adversaries [Unruh, DGK, CDGS]. The technique assumes a *single* preprocessing stage, and the reduction carrying a bit-fixing bound back to the auxiliary-input model [CDGS, Thm. 5] exploits that assumption directly: the bit-fixing attacker simulates the one pre-processor internally, on a table it samples itself, and can therefore inspect the whole of that table together with the whole of the advice before deciding which points to fix.

The conjecture below asks for the analogue with $\ell \geq 2$ preprocessing stages running in parallel: they read the same oracle, use independent private coins, and never communicate. The obstruction is not that any one leakage is long. It is that no bit-fixing

preprocessor can see the other stages' leakage, because that leakage depends on private coins it does not hold, and the points needing to be fixed are in general a function of all of them.

Two things follow from it, and are proved from it in [PP]. Any bound already proved in the *ordinary* bit-fixing model transfers to ℓ parallel preprocessing stages, so that existing bounds need not be reproved; and, running that transfer on one application, a random oracle extracts randomness from ℓ independent unpredictable sources — which is the application [CFHS, §1.4] raises the decomposition question for, compression being what stalls on the non-monolithic constructions a decomposition route is wanted for.

2 Notation and parameters

For $n \in \mathbb{N}$ write $[n] := \{1, \dots, n\}$. All logarithms are base two, and $x \leftarrow \mathcal{X}$ is a uniform draw from a finite set, independent of all others unless stated otherwise. Algorithms are computationally unbounded and may be randomized, so that “algorithm” may be read as “arbitrary, possibly randomized, function”; only query counts are ever bounded. An *absolute constant* does not depend on any parameter below. Indices i and j range over $[\ell]$.

Fix throughout:

- $\ell \in \mathbb{N}$ with $\ell \geq 2$, the number of parallel preprocessing stages.
- $N, M \in \mathbb{N}$, and $\text{Fun} := \{f : [N] \rightarrow [M]\}$, a function being identified with its table.
- $S_1, \dots, S_\ell \in \mathbb{N}$, the advice lengths of the stages.
- $P_1, \dots, P_\ell \in \mathbb{N}$, the numbers of points the stages may fix.
- $\gamma \in (0, 1)$, an additive slack.
- $\mathcal{R}_1, \dots, \mathcal{R}_\ell$, the private coin spaces.

Definition 1 (split preprocessing). A *split preprocessing* is a tuple of randomized maps $f_i : \text{Fun} \times \mathcal{R}_i \rightarrow \{0, 1\}^{S_i}$, one per stage, written $z_i = f_i(F; r_i)$. Each stage reads the whole of F and draws its own $r_i \leftarrow \mathcal{R}_i$; the coins $r_1, \dots, r_\ell$ are *mutually independent* and no stage sees another's coins or output. Write

$\mathbf{z} := (z_1, \dots, z_\ell)$ and $\mathbf{r} := (r_1, \dots, r_\ell)$.

This is the preprocessing of a split attacker against the *split auxiliary-input* oracle SAI-RO, which samples $F \leftarrow \text{Fun}$, hands it at the preprocessing interface separately to each of the ℓ parties, and answers the online interface by F itself. The conjecture's purpose is to replace that oracle by the *split bit-fixing* oracle SBF-RO, which instead accepts from party i a list of at most P_i query/answer pairs and answers online from the merged lists where they are defined and from a fresh table elsewhere. Neither model is needed to state the conjecture; both are set out in [PP, §3].

3 The conjecture

The object asked for is a tuple of *selectors*, one per stage, each reading the whole oracle but only its own leakage and its own coins.

Definition 2 (split selector tuples). A *split selector tuple* with budgets $(P_1, \dots, P_\ell)$ is a tuple of maps

$$\begin{aligned} \Lambda_i &: \text{Fun} \times \{0, 1\}^{S_i} \times \mathcal{R}_i \longrightarrow 2^{[N]} \times [M]^*, \\ \Lambda_i(F, z_i, r_i) &= (I_i, a_i), \end{aligned}$$

for $i \in [\ell]$, with $|I_i| \leq P_i$ and $a_i : I_i \rightarrow [M]$. The tuple is *consistent* if $a_i = F|_{I_i}$ for every i and all inputs, that is, if each selector fixes its points to the values the oracle actually takes there.

Given F , $\mathbf{z}$ and $\mathbf{r}$, write $I := \bigcup_i I_i$ and let $Y[\Lambda]$ denote the random function taking the value $(a_1 \triangleright \dots \triangleright a_\ell)[u]$ at each $u \in I$, where $\triangleright$ is the priority merge — at u it takes $a_i(u)$ for the least i with $u \in I_i$ — and an independent uniform value of $[M]$ at each $u \notin I$. Once $(F, \mathbf{z}, \mathbf{r})$ is fixed, $Y[\Lambda]$ is a P -bit-fixing source for $P := \sum_i P_i$, and $|I| \leq P$ always.

Under consistency the merge rule is immaterial, since for $u \in I_i \cap I_j$ both selectors fix u to $F(u)$; it is written into the definition only so that $Y[\Lambda]$ is defined for the free variants of the conjecture discussed in [PP, §6].

Conjecture 1 (parallel decomposition). *There is an absolute constant c such that the following holds for all $\ell \geq 2$, all N and M , all split preprocessings $f_1, \dots, f_\ell$ with output lengths $S_1, \dots, S_\ell$, all $P_1, \dots, P_\ell \in \mathbb{N}$ and all $\gamma \in (0, 1)$. There is a consistent split selector tuple $(\Lambda_1, \dots, \Lambda_\ell)$ with budgets $(P_1, \dots, P_\ell)$, depending only on $f_1, \dots, f_\ell, P_1, \dots, P_\ell$ and γ , such that, writing*

$$\Delta(T) := c T \sum_{i=1}^{\ell} \frac{S_i + \log \gamma^{-1}}{P_i},$$

the following two bounds hold for every $T \in \mathbb{N}$ and every distinguisher D that takes a tuple of strings and makes at most T oracle queries. Draw $X \leftarrow \text{Fun}$, then $r_1, \dots, r_\ell$ independently, set $z_i := f_i(X; r_i)$, and draw $Y[\Lambda]$ as in Definition 2 without resampling $X, \mathbf{z}$ or $\mathbf{r}$. Then

$$\left| \Pr[D^X(\mathbf{z}) = 1] - \Pr[D^{Y[\Lambda]}(\mathbf{z}) = 1] \right| \leq \Delta(T) + \gamma,$$

and, with $\Delta^+(T)$ denoting $\Delta(T)$ with each $\log \gamma^{-1}$ replaced by $2 \log \gamma^{-1}$,

$$\Pr[D^X(\mathbf{z}) = 1] \leq 2^{\Delta^+(T)} \cdot \Pr[D^{Y[\Lambda]}(\mathbf{z}) = 1] + 2\gamma.$$

4 What the quantification is doing

Five points are the statement rather than decoration. Each is developed, with proofs where there are any, in [PP].

Remark 1 (reduction to the known case). Taking f_j constant, so $S_j = 0$, and $P_j = 1$, for every $j \geq 2$, the statement is Lemma 1 of [CDGS] up to the constant c , with the single difference that f_1 is permitted to be randomized. That difference is free: the entropy bookkeeping the single-source proof rests on survives randomized leakage verbatim [PP, Lem. 1]. The entire content of the conjecture is that the fixed set may be presented as a union $I_1 \cup \dots \cup I_\ell$ whose parts are chosen without reference to one another's leakage.

Remark 2 (the selectors are split, and that is the difficulty). Running the single-source selector on each z_i separately and taking

the union fails already at $\ell = 2$. Identify $[M]$ with $\mathbb{Z}_M$, let f_1 draw $u \leftarrow [N]$ and output $(u, \sum_{x \neq u} X(x))$, let f_2 draw $v \leftarrow [N]$ and output $(v, \sum_{x \neq v} X(x))$, and take f_j constant for $j \geq 3$. Each leakage constrains X by one linear equation on $N - 1$ coordinates, so each conditional source is dense and the single-source selector fixes nothing; but given both, and $u \neq v$, the equations differ by $X(v) - X(u)$, and the two-query D that reads the oracle at u and at v — both named by its input — distinguishes with advantage $1 - 1/M$. Setting $I_1 := \{u\}$ and $I_2 := \{v\}$, each computable from its own leakage alone, repairs it. Whether that is a general phenomenon is exactly what is open. The example is [CFHS, §1.4]’s attack.

Remark 3 (why the selectors read the whole oracle). Λ_i is given F in full, and not merely z_i . That is what makes consistency achievable, and consistency is what makes the lists compatible where they overlap, so that no arbitration rule is needed and the statement is symmetric in the stages. It is also what the single-source construction already does. The selectors are *not* given the other stages’ leakage or coins, and that restriction, not the input F , is where the difficulty lies.

Remark 4 (quantifier order). The selector tuple is chosen after $P_1, \dots, P_\ell$ and γ but *before* T and before D . Relaxing this makes the statement easier in a way neither consequence in [PP] can use: both fix their query budget only after the attacker, and the extraction theorem there needs one tuple to serve two distinguishers at two different budgets.

Remark 5 (the shape of the error term is extrapolated). [CDGS] fixes $\Delta(T)$ only at $\ell = 1$. That the stages contribute a *sum* $\sum_i (S_i + \log \gamma^{-1})/P_i$, one term per stage of the single-source shape, is what one gets by charging each stage for its own leakage against its own budget and charging nothing for the interaction between them. Nothing establishes it beyond $\ell = 1$. A true statement with an extra factor $\phi(\ell)$, or with the budgets entering through $\min_i P_i$, would leave both consequences structurally intact and change every bound they produce. The constant c is the one thing this statement is casual about; the functional form is a second.

Remark 6 (unpredictable advice is not assumed, and marks the hard case). No hypothesis is placed on how guessable the z_i are,

deliberately. At one extreme, if each z_i is a deterministic function of the table the transfer is immediate from [CDGS] and needs no conjecture: one stage, which reads F , recomputes every other $z_j = f_j(F)$ itself, applies the single-source selector of [CDGS, Lem. 1] to $z_1 \parallel \dots \parallel z_\ell$ with budget P , and presets it, the others presetting nothing. Unpredictable advice therefore *strengthens* the challenge rather than easing it: it is exactly the regime in which no stage can reconstruct another's advice. Applications may supply such a hypothesis for their own reasons; the extraction theorem of [PP] does, and its hypothesis is about the *points* the advice encodes, not about the advice itself.

5 Bibliography

- [CDGS] S. Coretti, Y. Dodis, S. Guo, J. Steinberger. *Random Oracles and Non-Uniformity*. EUROCRYPT 2018; ePrint 2017/937.
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- [DGK] Y. Dodis, S. Guo, J. Katz. *Fixing Cracks in the Concrete: Random Oracles with Auxiliary Input, Revisited*. EUROCRYPT 2017.
- [PP] *Parallel Presampling: the conjecture, the transfer to the ordinary bit-fixing model, and multi-source extraction*. Conjura working document, `parallel-presampling.tex`, beside this statement. Contains the models, both consequences with proofs, the conflicting-list variants, the entropy bookkeeping, and the comparison with [CFHS].
- [Unruh] D. Unruh. *Random Oracles and Auxiliary Input*. CRYPTO 2007.